# Janu Kumar (J.K.) Sah

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## Research Interests

Thermal and fluid sciences · Fluid-structure interaction · Bio-inspired flow sensing · Computational mechanics · Sustainable energy and materials

## Education

**Ph.D., Mechanical Engineering** Jan 2026 – Present
*Rochester Institute of Technology, Kate Gleason College of Engineering* *Rochester, NY*

- **GPA:** 4.0/4.0
- **Research project:** *From Nature to Prototype: Development of a Bio-Inspired Seal Whisker Sensor Array*
- **Coursework:** Computational Methods in Fluid Dynamics, Continuum Mechanics, Advanced Engineering Mathematics, Engineering Analysis, Interdisciplinary Research Methods, Translating Discovery into Practice, Doctoral Seminar

**B.E., Mechanical Engineering** Nov 2018 – Apr 2023
*Pulchowk Campus, Institute of Engineering, Tribhuvan University* *Kathmandu, Nepal*

- **Overall score:** 76.36/100; merit-based full scholarship for undergraduate study.
- **Thesis:** Study of the energy scenario and improvement opportunities of the ICT building, Pulchowk Campus.

**English proficiency:** IELTS 7.0/9.0

## Technical Skills

- **Programming & Data:** Python, MATLAB; signal processing (FFT, spectral analysis); machine learning (neural operators).
- **Simulation & Modeling:** CFD and fluid-structure interaction (ANSYS CFX/Fluent, immersed-boundary methods); finite element method; reduced-order models.
- **Experiments & Instrumentation:** towing-tank testing, strain-gauge sensors, SPI sensor interfacing, ESP32 firmware, PCB debugging, oscilloscope / multimeter benchtop testing.
- **Thermal & Energy:** building and industrial energy audits; EnergyPlus, LEAP, PVsyst.
- **CAD:** SOLIDWORKS, AutoCAD, SketchUp (3D modeling and drafting).
- **General:** LaTeX, Git; experimental design, data analysis, and technical writing.

## Research Experience

**Graduate Research Assistant** Jan 2026 – Present
*Department of Mechanical Engineering, Rochester Institute of Technology* *Rochester, NY*

- Ran **towing-tank experiments** with a 9-sensor seal-whisker array on a moving carriage behind an independently towed cylinder (Re = 4000, 6000 and 8000); the measured shedding frequency scaled linearly with tow speed.
- Repaired a **32-channel sensor board** and rewrote its ESP32 firmware, recovering 14 of 16 whisker sensors.
- Assembled and ballasted a **BlueROV2** as the underwater test vehicle and ran its first towing-tank tests.

**Research Assistant** Apr 2023 – Mar 2024
*Center for Energy Studies, Institute of Engineering, Tribhuvan University* *Lalitpur, Nepal*

- Contributed to 25+ solar-monitoring projects and energy audits of industrial sites (Supervisor: Dr. Shree Raj Shakya).
- Research focus: national energy-intensity assessment, wind-energy potential for water pumping, and industrial sustainability-index calculation.

## Professional Experience

**Mechanical Design Engineer** Apr 2024 – Nov 2025
*Ozaki Design Office* *Niihama City, Japan*

- Performed CAD design, 3D modeling, and engineering simulation; conducted industrial site visits to study and analyze mechanical components.

**Intern** Apr – May 2022
*Alternative Energy Promotion Center* *Kathmandu, Nepal*

- Gained hands-on experience in energy auditing and management, renewable energy, and energy policy, regulation, and outreach.

## Publications

### Peer-Reviewed Journal Articles

1. **Sah, J. K.** (2026). When is carbonation-cured concrete actually cleaner? A systematic review, a net-benefit condition framework, and a proposed minimum reporting set for $CO_2$ mineralization in cement-based products. *Journal of Cleaner Production*, 579, 149605. doi:10.1016/j.jclepro.2026.149605

2. Shakya, S. R., **Sah, J. K.**, Bhujel, P., Aryal, H., Bajracharya, T. R., Bhandari, R., Subedi, S., & Chaulagain, B. (2024). Energy intensity improvement potential of the commercial buildings in developing countries: A case study of Nepal. *Journal of Innovations in Engineering Education*, 7(1), 84–95.
3. Shakya, S. R., Subedi, S., Bajracharya, T. R., Shrestha, S. L., **Sah, J. K.**, & Chaulagain, B. (2025). Sustainability Index for a Manufacturing Industry: A Case Study of Cement Industry in Nepal. *Journal of Science and Engineering*, 12(1), 37–48.
4. Shakya, S. R., Chaulagain, B., Poudel, S., Subedi, T. R., Poudel, P. S., Subedi, S., & **Sah, J. K.** (2025). Computation and Experimental Analysis of Helical Bach-Type Savonius Wind Turbine for Water Pumping Application. *Journal of the Institute of Engineering*, 18(1), 94–101.
5. Shah, M. P., & **Sah, J. K.** (2023). Design Optimization of Fins of a Sounding Rocket for Maximum Lift-to-Drag Ratio and Minimum Radar Cross-Section Area Using ANSYS. *International Journal on Engineering Technology*, 1(1), 53–61.

### Manuscripts Under Revision

1. **Sah, J. K.** Do neural operators beat classical solvers for linear elasticity? A strong-baseline, full-cost benchmark. *Computational Mechanics* (Springer), under revision.

### Peer-Reviewed Conference Proceedings

1. **Sah, J. K.**, Ghimire, I., Tamang, B., & Adhikari, N. (2025). CFX Analysis to Study the Effects of Blade Exit Angle on Performance of Centrifugal Pumps. In *Vibration Engineering and Technology of Machinery, Vol. I (VETOMAC 2022)*, Lecture Notes in Mechanical Engineering. Springer, Singapore.
2. **Sah, J. K.**, & Paudel, A. (2023). Cost Analysis of Owning Four-Wheel Electric Vehicles in Nepal. *International Conference on Engineering & Technology*, Vol. 5, KEC, 112–116.

## Selected Projects

- **Industrial Energy Audit, Dairy Development Corporation, Kathmandu.** Proposed an integrated recovery system: 138 KVA/month demand reduction via power-factor correction (0.67→0.98) and 15% boiler-fuel savings through insulated cladding (2-month payback).
- **EV Cost-Competitiveness Model (Kathmandu Valley).** MATLAB lifetime-cost model showing 40% lower cost for converted EVs vs. new ICE vehicles at 10,000 km/yr.

## Talks and Presentations

- **Oral presentation:** "From Nature to Prototype: Development of a Bio-Inspired Seal Whisker Sensor Array," 2026 RIT Graduate Showcase. Apr 2026
- **Exhibitor:** "From Nature to Prototype: Seal Whisker Sensors," Imagine RIT. Apr 2026
- **Poster:** "Effects of blade exit angle on the performance of a centrifugal pump," VETOMAC 2022, Tribhuvan University, Nepal. Dec 2022

## Honors and Awards

RIT Ph.D. Merit Scholarship / Graduate Assistantship — Jan 2026 – Present

Merit-Based Full Scholarship for Undergraduate Study — Nov 2018 – Apr 2023

## Professional Service, Leadership & Outreach

- **Reviewer**, *International Journal of Concrete Structures and Materials* (Springer).
- **Reviewer**, *Journal of Building Engineering* (Elsevier).
- **President**, Saraswati Pooja Committee, NTBNS, IOE Pulchowk Campus (2022–2023).
- **Exhibition Coordinator**, 13th National Mechanical and Aerospace Engineering Exhibition, SOMAES (2022–2023).